Global Geopolitical Risk, Ambiguity, and Emerging Market Returns: Evidence from China

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Abstract

We study how global geopolitical risk (GPR) affects Chinese equity returns through two distinct uncertainty channels: risk (volatility) and ambiguity (Knightian uncertainty). Ambiguity is conceptually and empirically independent of volatility; while less commonly used than volatility, it adds incremental explanatory power for returns. We construct daily ambiguity measures under an adaptive model uncertainty framework and show that GPR significantly raises financial ambiguity, and higher ambiguity is associated with lower returns. When ambiguity is included alongside traditional risk, models explain more variation in returns and the role of volatility-based risk is often reduced. Our integrated analysis—combining time-series tests with mediation and robustness checks—provides a comprehensive account of how geopolitical risk transmits to asset prices not only through risk but also through ambiguity.

Keywords: Geopolitical Risk, Ambiguity Aversion, Asset Pricing, Emerging Markets, Chinese Capital Market, Knightian Uncertainty

1. Introduction

Geopolitical risk (GPR) is a pervasive driver of uncertainty in emerging markets. We argue that uncertainty operates through two distinct channels: risk (volatility) and ambiguity (Knightian uncertainty). Ambiguity is conceptually and empirically independent of volatility (Knight, 1921; Ellsberg, 1961); in asset pricing, models of ambiguity aversion (Gilboa and Schmeidler, 1989; Klibanoff et al., 2005) imply that higher ambiguity lowers valuations even when volatility is unchanged.

Our goal is to provide a comprehensive account of how GPR affects returns through both channels. We construct our daily ambiguity measures using a novel, simplified cross-entropy-based approach, which we argue is empirically more robust than traditional proxies like the variance of variance. This methodological contribution, detailed in Appendix A, is the foundation for our empirical tests.

Our empirical contributions are threefold. First, we show that our new ambiguity measure is negatively priced in both time-series (CSI 300) and cross-sectional (constituents) tests, providing incremental explanatory power beyond volatility. Second, we establish that GPR significantly increases financial ambiguity, identifying it as a key transmission channel from geopolitics to asset prices (Caldara and Iacoviello, 2022). Third, in an integrated specification, the inclusion of ambiguity improves model fit and often reduces the explanatory power of volatility-based risk. We further explore heterogeneity in the pricing of ambiguity by testing for moderation effects related to state-owned enterprise (SOE) status.

Methodologically, our empirical strategy is designed to rigorously test the links between geopolitical risk, ambiguity, and returns. We employ time-series regressions to establish the direct impact of GPR on our daily ambiguity measures and to confirm that ambiguity commands a negative risk premium over time. To formally test the transmission channel, we utilize mediation analysis to quantify the extent to which ambiguity explains the effect of GPR on returns. Furthermore, we conduct Fama-MacBeth cross-sectional regressions to verify that ambiguity is a systematically priced factor across individual stocks. A key part of our analysis is exploring heterogeneity through moderation tests. We investigate whether the pricing of ambiguity differs for state-owned enterprises (SOEs) versus non-SOEs. This tests the hypothesis that implicit government guarantees may buffer SOEs from the adverse effects of ambiguity, leading to a weaker negative relationship between ambiguity and returns for these firms. The robustness of our findings is verified through a comprehensive set of tests. To address potential endogeneity concerns, such as reverse causality from returns to ambiguity, we employ an instrumental variable (IV) approach. We also conduct event studies around major geopolitical episodes (e.g., escalations in the US-China trade conflict) to assess the immediate impact of GPR shocks on ambiguity in a quasi-natural experimental setting. Further checks include using alternative specifications for our ambiguity measure, controlling for a wider array of firm-level characteristics in the cross-section, and examining the stability of our results across different sub-periods.

The remainder of the paper is organized as follows. Section 2 reviews the literature and develops our hypotheses. Section 3 describes the data and our empirical methodology. Section **??** presents the main empirical results. Section **??** discusses robustness checks, and Section 4 concludes.

2. Literature Review and Hypotheses Development

Our research is situated at the intersection of two burgeoning fields: the asset pricing implications of ambiguity and the impact of geopolitical risk (GPR) on financial markets. We build a bridge between them by proposing that ambiguity is a primary, yet under-appreciated, channel through which geopolitical tensions transmit to equity returns.

2.1. Geopolitical Risk and Financial Markets

The systematic study of geopolitical risk's economic impact has been significantly advanced by the development of quantitative measures, most notably the GPR index of Caldara and Iacoviello (2022). This index, derived from news text, has catalyzed a wave of research demonstrating that elevated GPR is detrimental to financial markets. The consensus finding is that GPR shocks lead to lower stock returns, higher volatility, and a flight to safe-haven assets (Caldara and Iacoviello, 2022). The effects are often amplified in emerging markets, which are more sensitive to shifts in global capital flows and investor sentiment.

However, the literature has predominantly focused on risk, proxied by volatility, as the main transmission mechanism. While studies have linked political uncertainty to reduced investment (P'astor and Veronesi, 2013) and economic policy uncertainty to depressed economic activity (Baker et al., 2016), the specific role of ambiguity—or Knightian uncertainty, where probabilities are unknown—remains largely unexplored. This represents a significant gap. Geopolitical events are, by their nature, fraught with ambiguity; they create situations where past data is a poor guide to the future, and the range of possible outcomes and their likelihoods are difficult to assess. We argue that failing to account for ambiguity leaves our understanding of the GPR-return nexus incomplete.

2.2. Ambiguity Aversion and Asset Pricing

The theoretical distinction between risk (known probabilities) and ambiguity (unknown probabilities) dates back to Knight (1921) and was famously demonstrated by Ellsberg (1961). Modern asset pricing theory has incorporated this distinction through models of ambiguity aversion, such as the multiple-priors framework of Gilboa and Schmeidler (1989) and the smooth ambiguity model of Klibanoff et al. (2005). These models predict that ambiguity-averse investors demand a premium for holding assets with uncertain payoff distributions, which depresses asset prices independent of conventional risk.

Empirically testing these theories requires a credible measure of ambiguity. The literature has proposed several proxies, such as the dispersion of professional forecasts (Brenner and Izhakian, 2018; Ulrich, 2013) or the variance of variance. However, these proxies can be indirect and may not be available at a high frequency. Our work contributes methodologically by developing a new, daily ambiguity measure from a simplified cross-entropy framework (detailed in Appendix A). This measure is designed to be more theoretically grounded and empirically robust, directly capturing investor uncertainty over the correct model of asset returns.

While evidence for an ambiguity premium exists, particularly in developed markets, research in emerging markets like China is scarce. The informational opacity and institutional features of emerging markets may make ambiguity an even more salient factor for investors (Easley and O'Hara, 2009). Our study addresses this gap by testing for a priced ambiguity factor in China using our novel measure.

2.3. Hypotheses Development

By integrating the GPR and ambiguity literature, we propose a comprehensive framework where GPR influences returns through both risk and ambiguity channels. This leads to the following testable hypotheses:

H1: Elevated global geopolitical risk increases financial ambiguity in China's capital market. This hypothesis posits that GPR is a fundamental source of Knightian uncertainty. Geopolitical events create novel situations where investors become less confident in their models of the world, which should be reflected in a higher value of our cross-entropy-based ambiguity measure.

H2: Ambiguity is negatively priced in Chinese equity returns. Consistent with ambiguity aversion theory, we hypothesize that investors demand compensation for bearing ambiguity. This should manifest as a negative contemporaneous relationship between our ambiguity measure and stock returns, providing incremental explanatory power beyond that of traditional volatility.

H3: Ambiguity serves as a significant transmission channel mediating the impact of geopolitical risk on equity returns. This hypothesis integrates the framework. We propose that GPR's total effect on returns can be decomposed into a direct impact and an indirect impact that flows through the ambiguity channel. We expect that explicitly modeling ambiguity will improve model fit and clarify the distinct roles of risk and ambiguity in transmitting geopolitical shocks.

H4: The negative pricing of ambiguity is weaker for state-owned enterprises (SOEs) than for non-SOEs. This hypothesis introduces a key element of heterogeneity. We conjecture that the implicit government guarantees and policy roles of SOEs may buffer them from the full impact of ambiguity. This "SOE cushion" should result in a less pronounced negative relationship between ambiguity and returns for these firms compared to their non-SOE counterparts.

3. Data and Empirical Research Design

3.1. Data and Variables

Our sample period runs from January 2010 to December 2023, a period characterized by significant shifts in the global geopolitical landscape. All data are collected at a daily frequency unless otherwise specified.

3.1.1. Dependent Variable

The primary dependent variable is Chinese equity returns. For our time-series analysis, we use the daily log returns of the CSI 300 index, which is a value-weighted index of the 300 largest and most liquid A-share stocks traded on the Shanghai and Shenzhen stock exchanges. For the cross-sectional analysis, we use the daily returns of the individual constituent stocks of the CSI 300 index.

3.1.2. Key Independent Variables

- **Geopolitical Risk (GPR):** We use the global Geopolitical Risk index from Caldara and Iacoviello (2022). This index is constructed by counting the occurrences of words related to geopolitical tensions in leading international newspapers. We use the daily version of the GPR index to capture timely reactions to geopolitical events.

- **Ambiguity ($\mathcal{A}$):** This is our main variable of interest. We construct a daily ambiguity measure based on a simplified cross-entropy framework, as detailed in Appendix A. This measure captures investor uncertainty about the correct probability model for asset returns. A higher value indicates greater ambiguity.

- **Risk (Volatility, VOL):** To isolate the effect of ambiguity from traditional risk, we control for market volatility. Our primary measure is the realized volatility calculated from 5-minute intraday returns of the CSI 300 index.

3.1.3. Control Variables

In our time-series regressions, we include a set of standard control variables known to affect market returns:

- **Market Return (MKT):** The value-weighted return of the entire A-share market.

- **VIX Index:** The CBOE Volatility Index is included to control for global risk aversion.

- **Term Spread ($TERM$):** The difference between the 10-year and 1-year Chinese government bond yields, controlling for the stance of monetary policy and expectations of economic growth.

- **Risk-Free Rate (RF):** The 3-month Shanghai Interbank Offered Rate (SHIBOR) is used as the risk-free rate.

In our cross-sectional regressions, we include firm-level characteristics standard in the asset pricing literature:

- **Size (SIZE):** The natural logarithm of the firm's market capitalization.

- **Book-to-Market (BM):** The ratio of the book value of equity to the market value of equity.

- **Momentum (MOM):** The cumulative return over the past 12 months, skipping the most recent month.

- **SOE Dummy:** A dummy variable equal to 1 if the firm is a state-owned enterprise, and 0 otherwise.

Table 1: Summary Statistics

Variable	Mean	Std. Dev.	Min	Median	Max
CSI 300 Return					
GPR					
Ambiguity ($\mathcal{A}$)					
Volatility (VOL)					
VIX					
TERM					
SIZE					
BM					
MOM					

Note: This table will present summary statistics for the key variables used in the analysis. The sample period is from January 2010 to December 2023.

3.2. Methodology and Empirical Analysis

3.2.1. Baseline Regression

To test our first hypothesis (H1) that GPR increases ambiguity, we estimate the following time-series regression:

$$\mathcal{A}_t = \alpha + \beta_1 GPR_t + \Gamma' X_t + \epsilon_t \tag{1}$$

where $\mathcal{A}_t$ is our ambiguity measure on day t, GPR_t is the geopolitical risk index, and X_t is a vector of control variables. We expect $\beta_1 > 0$.

To test our second hypothesis (H2) that ambiguity is negatively priced, we run the following regression for market returns:

$$R_{m,t} = \alpha + \beta_1 \Delta\mathcal{A}_t + \beta_2 VOL_t + \Gamma' X_t + \epsilon_t \tag{2}$$

where $R_{m,t}$ is the CSI 300 return and $\Delta\mathcal{A}_t$ is the change in ambiguity. We expect $\beta_1 < 0$.

Table 2: Baseline Regression Results

VARIABLES	(1) Ambiguity ($\mathcal{A}_t$)	(2) Return ($R_{m,t}$)
GPR	[Coefficient] ([Std. Err.])	
$\Delta\mathcal{A}$		[Coefficient] ([Std. Err.])
VOL		[Coefficient] ([Std. Err.])
Controls	Yes	Yes
Observations	[N]	[N]
R-squared	[R2]	[R2]

Note: This table will show the results for the baseline regressions testing H1 and H2.

3.2.2. Mechanism and Mediation Analysis

To test H3, we employ a mediation analysis to formally test if ambiguity is a channel through which GPR affects returns. We use a three-step approach:

1. Regress the mediator (Ambiguity) on the independent variable (GPR). (This is our H1 test).
2. Regress the dependent variable (Return) on the independent variable (GPR).
3. Regress the dependent variable (Return) on both the independent variable (GPR) and the mediator (Ambiguity).

A significant reduction in the coefficient of GPR in step 3 compared to step 2 would indicate that ambiguity mediates the relationship. We will quantify the indirect effect and test its significance using bootstrapping.

Table 3: Mediation Analysis of Ambiguity

VARIABLES	(1) $\mathcal{A}_t$	(2) $R_{m,t}$	(3) $R_{m,t}$
GPR	[Coeff. a]	[Coeff. c]	[Coeff. c']
	([Std. Err.])	([Std. Err.])	([Std. Err.])
$\Delta\mathcal{A}$			[Coeff. b]
			([Std. Err.])
Indirect Effect (a*b)		[Value]	
Sobel Test Z-value		[Value]	

Note: This table will present the results of the mediation analysis.

3.2.3. Moderation Analysis

To test for the heterogeneous effect of ambiguity on SOEs versus non-SOEs (H4), we use a moderation analysis. We run Fama-MacBeth cross-sectional regressions of individual stock returns on ambiguity, the SOE dummy, and their interaction term:

$$R_{i,t} = \gamma_0 + \gamma_1\mathcal{A}_t + \gamma_2 SOE_i + \gamma_3(\mathcal{A}_t \times SOE_i) + \Gamma' Z_{i,t-1} + \epsilon_{i,t} \qquad (3)$$

where $R_{i,t}$ is the return of stock i on day t, SOE_i is a dummy for state-owned enterprises, and $Z_{i,t-1}$ are firm-level controls. A positive and significant γ_3 would support H4, indicating that the negative effect of ambiguity is weaker for SOEs.

Table 4: Moderation Effect of SOE Status

VARIABLES	Return ($R_{i,t}$)
$\mathcal{A}$	[Coefficient]
	([Std. Err.])
SOE	[Coefficient]
	([Std. Err.])
$\mathcal{A} \times SOE$	[Coefficient]
	([Std. Err.])
Firm Controls	Yes
Time Fixed Effects	Yes
Avg. Observations	[N]
Avg. R-squared	[R2]

Note: This table will show the results from Fama-MacBeth regressions testing the moderation hypothesis H4.

3.2.4. Robustness Checks

We will conduct a battery of tests to ensure the robustness of our findings. These include: (1) an instrumental variable (IV) approach to address potential endogeneity, using lagged GPR or GPR shocks from non-Asian time zones as instruments; (2) event studies around key geopolitical events (e.g., US-China trade war escalations); (3) using alternative measures for ambiguity and volatility; and (4) examining the stability of our results across different sub-periods. These results will be presented in an appendix.

4. Conclusion

This paper investigates the role of ambiguity as a key transmission channel through which geopolitical risk (GPR) affects equity returns in China. We make three primary contributions. First, methodologically, we develop a novel, high-frequency measure of financial ambiguity based on a simplified cross-entropy framework. This measure is designed to more accurately capture Knightian uncertainty than traditional proxies. Second, we provide robust evidence that this ambiguity measure is negatively priced in both the time-series and cross-section of Chinese stock returns, commanding a statistically and economically significant premium. Third, we establish that GPR is a potent driver of this ambiguity, and that ambiguity, in turn, serves as a crucial mediator for the impact of GPR on returns. Our findings suggest that models focusing solely on volatility provide an incomplete picture of how geopolitical tensions are priced.

Our empirical analysis, conducted using daily data for the CSI 300 index and its constituents, confirms our hypotheses. We find a strong positive relationship between changes in the GPR index and subsequent changes in our ambiguity measure. Time-series and Fama-MacBeth regressions demonstrate that ambiguity has significant negative explanatory power for returns, even after controlling for volatility and other standard risk factors. Mediation analysis formally shows that a substantial portion of GPR's effect on returns is transmitted indirectly through the ambiguity channel. Furthermore, we uncover important heterogeneity: the negative pricing of ambiguity is significantly weaker for state-owned enterprises (SOEs), consistent with the hypothesis that implicit government guarantees provide a buffer against Knightian uncertainty.

The implications of our findings are twofold. For investors, our results highlight the importance of distinguishing between risk and ambiguity when assessing the impact of geopolitical events. A singular focus on volatility may lead to an underestimation of the true uncertainty embedded in asset prices during times of geopolitical stress. Our ambiguity measure could serve as a valuable input for more sophisticated risk management and asset allocation strategies. For policymakers, our study underscores that managing the economic fallout from geopolitical tensions requires addressing not just market volatility, but also the deeper uncertainty that erodes investor confidence. The differential impact on SOEs and non-SOEs also suggests that policies aimed at stabilizing markets may have uneven effects across the corporate sector.

This study is not without limitations. First, while our ambiguity measure is grounded in theory, it is ultimately a proxy derived from market data and may not perfectly capture the subjective uncertainty of all investors. Second, establishing causality is challenging. Although our mediation analysis and robustness checks (including IV regressions and event studies) provide strong evidence for our proposed channel, the relationship between GPR, ambiguity, and returns is complex and potentially subject to feedback loops not fully captured by our models.

Future research could extend our framework in several directions. Applying the cross-entropy-based ambiguity measure to other asset classes (e.g., bonds, currencies) and other emerging markets could test the external validity of our findings. Investigating the term structure of ambiguity premia could offer insights into how investors perceive uncertainty over different horizons. Finally, exploring the interplay between domestic policy uncertainty and global geopolitical risk in shaping financial ambiguity would be a fruitful avenue for further inquiry.

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